TASK 2:

1) import Foundation

```
let itemPrice = 450.0

let quantity = 3

let subtotal = itemPrice * Double(quantity)

let hasCoupon = true

let discount = (hasCoupon && subtotal > 1000) ? subtotal * 0.10 : 0.0

let discountedAmount = subtotal - discount

let gst = discountedAmount * 0.18

let total = discountedAmount + gst

let numberOfPeople = 4

let share = total / Double(numberOfPeople)

let amount = 1234

let hundreds = amount / 100

let remainder = amount % 100

print("Item Price: ₹\(itemPrice)")

print("Quantity: \(quantity)")

print("Subtotal: ₹\(subtotal)")

print("Coupon Available: \(hasCoupon)")

print("Discount: ₹\(discount)")

print("GST: ₹\(gst)")

print("Total Bill: ₹\(total)")

print("Each Person's Share: ₹\(String(format: "%.2f", share))")

print("Hundreds: \(hundreds)")

print("Remainder: ₹\(remainder)")
```

OUTPUT:

Output

```
Item Price: ₹450.0
Quantity: 3
Subtotal: ₹1350.0
Coupon Available: true
Discount: ₹135.0
GST: ₹218.7
Total Bill: ₹1433.7
Each Person's Share: ₹358.43
Hundreds: 12
Remainder: ₹34
```

2)

```
let year1 = 1900
let year2 = 2000
let year3 = 2024
let year4 = 2026
let leap1 = (year1 % 4 == 0 && year1 % 100 != 0) || year1 % 400 == 0
let leap2 = (year2 % 4 == 0 && year2 % 100 != 0) || year2 % 400 == 0
let leap3 = (year3 % 4 == 0 && year3 % 100 != 0) || year3 % 400 == 0
let leap4 = (year4 % 4 == 0 && year4 % 100 != 0) || year4 % 400 == 0
print("1900 is a leap year: \(leap1)")
print("2000 is a leap year: \(leap2)")
print("2024 is a leap year: \(leap3)")
print("2026 is a leap year: \(leap4)")
let number = 75
let isInRange = (1...100).contains(number)
print("Number is between 1 and 100: \(isInRange)")
let value = 24
let isEvenNotMultipleOf10 = (value % 2 == 0) && (value % 10 != 0)
print("Even and not a multiple of 10: \(isEvenNotMultipleOf10)")
let nickname: String? = nil
let displayName = nickname ?? "Guest"
print("Nickname: \(displayName)")
```

OUTPUT:

```
Output

1900 is a leap year: false
2000 is a leap year: true
2024 is a leap year: true
2026 is a leap year: false
Number is between 1 and 100: true
Even and not a multiple of 10: true
Nickname: Guest
```

3) print(7 / 2)

print(-7 % 3)

print(7.0 / 2)

print(10 - 2 * 3 + 4 / 2)

print(5 > 3 && 2 > 4 || true)

print(1 << 3)

print(6 & 3, 6 | 3, 6 ^ 3)

let x: UInt8 = 255

print(x &+ 1)

OUTPUT:

```
Output

3
-1
3.5
6
true
8
2 7 5
0
```